



Activation of peroxymonosulfate by single-atom Fe-g-C₃N₄ catalysts for high efficiency degradation of tetracycline via nonradical pathways: Role of high-valent iron-oxo species and Fe–N_x sites

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ABSTRACT

A Fenton-like catalyst was prepared based on isolated Fe single-atom and Fe clusters anchored onto a g-C₃N₄ framework. It exhibited high activity and stability in the heterogeneous activation of peroxymonosulfate (PMS) for tetracycline (TC) degradation. Both experimental and density functional theory calculation results demonstrated that the unique N-coordinated single Fe atom (Fe–N₄) served as active sites with optimal binding energy for PMS activation. The experimental analysis indicated that the single Fe atoms displayed superior catalytic activity to Fe clusters. Furthermore, both high-valent iron-oxo species and single oxygen-dominated nonradical processes caused TC degradation. Among them, single Fe atoms anchored onto Fe-g-C₃N₄ served as Fe–N₄ reactive sites can directly activate PMS to produce high-valent iron-oxo species, which was the key active nonradical species causing TC degradation.

1. Introduction

Antibiotics have been considered a hazardous organic contaminant over the past few decades [1]. Yearly, 30%–90% of antibiotics are released into the environment, and significant concentrations have even been detected in freshwater [2]. Excessive discharge of these toxic antibiotics inhibits the growth of aquatic organisms and produces antibiotic-resistant bacteria, threatening human health and the sustainable development of society [3]. Therefore, increasing attention has been paid to the degradation of these refractory pollutants in the water environment. Advanced oxidation processes and generating numerous high reactive oxygen species (ROS), such as hydroxyl radical (•OH), sulfate radical (SO₄^{•−}), superoxide radical (O₂^{•−}), and singlet oxygen (¹O₂) can be used for the efficient degradation of organic compounds in water and wastewater [4]. Among these, Fenton-like reactions based on peroxymonosulfate (PMS) have shown enormous potential recently [5–8].

Various PMS activation methods, including ultraviolet, carbon catalysts, transition metals, and so on have been reported [9,58]. For the transition metals, Co, Fe, or Mn are the most effective activators for PMS

[10]. Especially, Fe is the most abundant transition metal in the earth's crust, which is characterized by high activity, relatively low cost, and environmentally friendly [11]. The nonradicals, including high-valent iron-oxo species, can also be generated when Fe-based catalysts were used as a heterogeneous PMS activator [12]. Peng et al., [13] calcined Enteromorpha biomass directly to obtain Fe–N–C catalysts. To prove the formation of high-valent iron-oxo species, dimethyl sulfoxide (DMSO) was added, and the products were analyzed using gas chromatography mass spectrometry (GC–MS) when activating PMS to degrade paracetamol. Electron paramagnetic resonance (EPR), quenching experiments, and electrochemical analysis confirmed that high-valent iron-oxo species were the main ROS for paracetamol degradation in the Fe–N–C/PMS system. Li et al., [12] also proposed that the degradation of bisphenol A (BPA) in PMS activated by Fe-based nanoparticles-loaded N-doped carbon (Fe NP–N–C) was realized by high-valent iron-oxo species. Therefore, Fe-based catalysts have potential application prospects in activating PMS to degrade organic pollutants.

In transition metal and N co-doped carbon materials, the electron-rich N dopants can create Lewis basic sites, which are prone to combine with the transition metal (Lewis acid) to form a strong M–N (M

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refers to metal Fe, Co, etc.) coordination in the carbon matrix (M–N–C) [14,56]. The N element has a higher electronegativity than the carbon element, which can enhance the oxidation efficiency of M–N–C materials in N and metal-containing carbon materials [15]. However, current synthesis methods in fulfilling high-metal loading occur through metal nanoparticles or atomic clusters forming during pyrolysis and reaction processes. Because of the influence of acidic conditions or agglomeration, the metal use efficiency is low, and the catalysts' performance is unideal [12]. Single-atom catalysts (SACs), a catalyst with monodisperse metal atoms anchored onto supports, have shown high efficiency and excellent selectivity in environmental catalysis, far better than metal nanoparticle catalysts [16,17,57]. Fe-based SACs are Fe–N_x sites in the Fe–N–C framework [18]. Fe–N_x sites, formed by the coordination of a single Fe atom with N, are served as active sites and realized the maximum use of metal sites that improve the catalytic efficiency [19]. However, the isolated single Fe atoms are easy to migrate and agglomerate to form nanoparticles in the pyrolysis process, decreasing Fe–N_x active center catalytic performance [20,21]. Therefore, the stability of single Fe atom catalysts must be further explored. The synthesis of single-atom Fe–g–C₃N₄ catalysts and the ability of the Fe atom to activate the H₂O₂ oxidant have been reported in [22–24], but the catalytic mechanism and the role of the single Fe atom in activating the oxidant remain unclear. Furthermore, the roles of the single atom and clusters were not identified.

In this study, a single Fe atom catalyst (SA Fe–g–C₃N₄) was prepared using Fe imidazole coordination compound (Fe ICC) as a precursor and g–C₃N₄ as support. This study obtained deeper insights into the performance of SA Fe–g–C₃N₄ as PMS activator for tetracycline (TC) in Fenton-like catalytic oxidation reactions and evaluated the roles of intrinsic Fe and N in the catalytic process. Furthermore, the dominant responsibility of a single atom and clusters in the SA Fe–g–C₃N₄ catalyst for TC

degradation via the PMS catalysis system was analyzed. To obtain a more atomic-level dispersion of Fe–N_x active sites, the formation energies of different coordination configurations were calculated using density functional theory (DFT) calculations compared with previous studies. The influence of different coordination forms of Fe and N atoms in SA Fe–g–C₃N₄ catalysts on different Fe–N_x coordination structures was determined. This study provides insights into the potential mechanisms for accelerated TC degradation, providing a new idea on the design of a single-atom Fe-based catalyst for TC degradation in water.

2. Experimental

2.1. Materials

Polyvinylpyrrolidone (PVP), 2-methylimidazole (2-MI), melamine, methanol (MeOH), iron (II) sulfate heptahydrate (FeSO₄·7H₂O), ethanol (EtOH), ferrous ferric oxide (Fe₃O₄), potassium iodide (KI), furfuryl alcohol (FFA), isopropyl alcohol (IPA), *p*-Benzoquinone (*p*-BQ), *tert*-butanol (TBA), citrate, oxalate, sodium sulfocyanate (NaSCN), dimethyl sulfoxide (DMSO), tetracycline (TC), methylene blue (MB), ciprofloxacin (CIP), norfloxacin (NOR), bright spring red 5R (BP-5R) and bisphenol A (BPA) were of analytical grade. Sources of chemicals are given in the Supporting Information.

2.2. Synthesis of catalysts

The single Fe-atom catalyst was synthesized according to the previous report with some modifications [22] (see Fig. 1a). Briefly, 0.172 M of 2-MI, 0.117 M of PVP and 0.036 M of FeSO₄·7H₂O were added into a zirconia milling jars, and then blended at 480 rpm for 2 h. The obtained dark green solid was centrifuged and washed 5 times with ethanol,

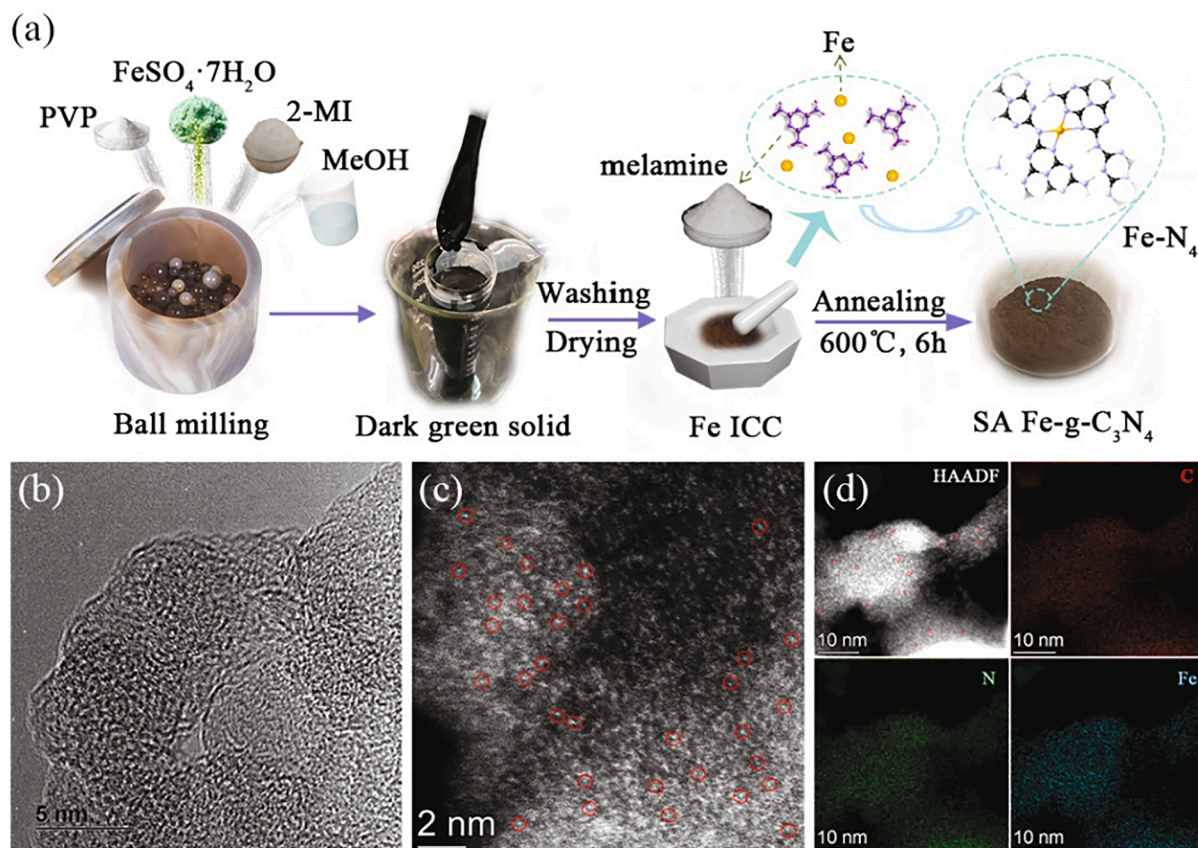


Fig. 1. (a) Schematic illustration of the preparation of SA Fe–g–C₃N₄ catalysts; (b) STEM image, (c) AC-HAADF-STEM images and the (d) corresponding EDX elemental mappings of SA Fe–g–C₃N₄ (600) catalysts.

followed by drying in vacuum at 60 °C for 12 h. The obtained product (Fe ICC) was mixed with melamine in a definite weight ratio (melamine/Fe ICC = 5), fully ground in an agate mortar and then carbonized at different pyrolysis temperatures for 6 h in a tube furnace under an N₂ atmosphere. The resulting catalysts pyrolyzed at 400, 500 and 600 °C were denoted as SA Fe-g-C₃N₄(400), SA Fe-g-C₃N₄(500) and SA Fe-g-C₃N₄(600), respectively. For comparison, melamine was directly calcined use the same method to obtain g-C₃N₄, and SA Fe-g-C₃N₄(600) catalyst was mixed with a 5 M H₂SO₄ at 80 °C for 4 h to remove metal clusters, then washed with deionized water until pH reached neutral. The characterization methods for the SA Fe-g-C₃N₄ catalysts were detailed in the [Supplementary Information](#).

3. Results and discussion

3.1. Characterization of SA Fe-g-C₃N₄

Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) were used to investigate the morphology and microstructures of prepared catalysts. [Fig. 1b](#) and [Fig. S1](#) show that the morphology of SA Fe-g-C₃N₄ (600) and SA Fe-g-C₃N₄ (500) presented a massive shape with a smooth surface, and no obvious aggregation of Fe nanoparticles could be observed. However, a small number of nanoparticles were found in SA Fe-g-C₃N₄ (400). The aberration-corrected (AC) high-angle annular dark-field (HAADF) scanning TEM (STEM) (AC-HAADF-STEM) image ([Fig. 1c](#) and [Fig. S2](#)) further confirms the absence of agglomerated Fe nanoparticles in SA Fe-g-C₃N₄ (600), and the uniformly dispersed bright spots less than 2 Å can be observed. Furthermore, a few atomic clusters were anchored onto the g-C₃N₄ framework. The energy dispersive X-ray element mapping ([Fig. 1d](#)) analysis confirmed that C, N, and Fe species were homogeneously

distributed on the catalyst. As shown in [Fig. 2a](#), the XRD analysis indicates that one diffraction peak at ~ 27.6° (2θ) was observed in the SA Fe-g-C₃N₄ (500 °C and 600 °C), and the absence of metallic Fe diffraction peaks could be detected for SA Fe-g-C₃N₄ catalysts (500 °C and 600 °C), indicating that Fe species are highly dispersed as single Fe atoms or tiny Fe clusters, which was consistent with the XAS and AC-HAADF-STEM results [25].

Raman spectroscopy was conducted to investigate the surface defect states of the synthesized catalysts. In [Fig. 2b](#), two distinct peaks, which were attributed to the D and G bands, were found in all samples at ~ 1349 cm⁻¹ and 1578 cm⁻¹, respectively. The I_D/I_G relative intensity ratios of the D band to the G band increased from 1.04 to 1.07 and 1.29 with increasing pyrolysis temperature. Therefore, the defect density of synthesized catalysts, including vacancies, edges, and topologic defects, can be increased with the increase in pyrolysis temperature because of the reconstruction of the carbon substrate in the reduction process of Fe species, resulting in a slightly increased defective degree [26].

The chemical composition of SA Fe-g-C₃N₄ was further studied via XPS. As presented in [Fig. 2c](#), C (287.9 eV, 284.8 eV), N (398.8 eV), O (531.7 eV), and Fe (710.7 eV, 724.4 eV) were detected in SA Fe-g-C₃N₄ (600), (500), and (400) catalysts, respectively. Table S1 shows the mass percentages of the as-prepared catalysts. The relative mass percentages of Fe in the obtained catalyst increased from 0.76 wt% to 4.07 wt% when the pyrolysis temperature was increased from 400 °C to 600 °C, which significantly increased by 5.4 times. The Fe 2p peak could be observed in SA Fe-g-C₃N₄(400), (500), and (600) catalysts ([Fig. 2d](#)). These results show that Fe was successfully incorporated into g-C₃N₄ and a higher Fe content can be obtained at 600 °C. The high-resolution XPS N1s curve of SA Fe-g-C₃N₄ catalysts ([Fig. S3](#)) revealed four peaks at 404.2, 400.8, 399.7, and 398.5 eV, corresponding to nitrogen oxides, graphitic N, pyrrolic N, and pyridinic N, respectively

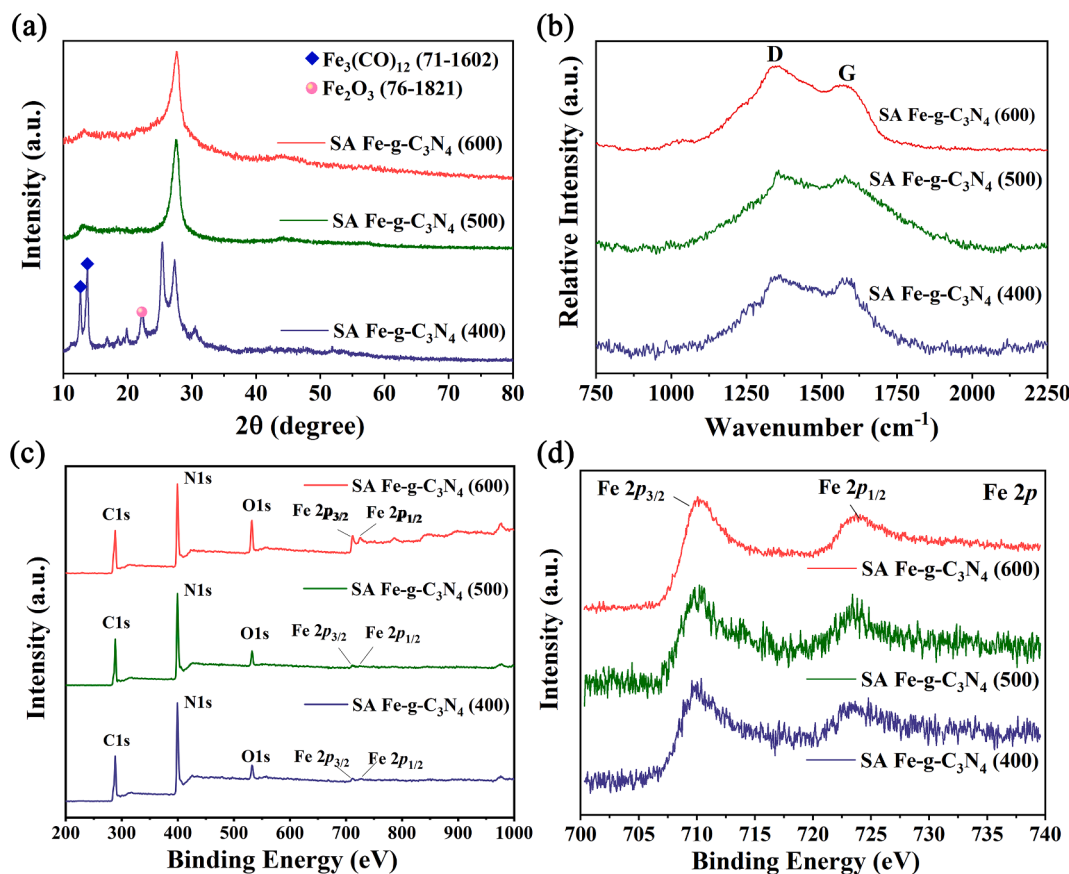


Fig. 2. (a) XRD and (b) Raman spectra of SA Fe-g-C₃N₄(600), SA Fe-g-C₃N₄(500) and SA Fe-g-C₃N₄(400); (c-d) XPS spectra of SA Fe-g-C₃N₄ (400), SA Fe-g-C₃N₄ (500) and SA Fe-g-C₃N₄ (600) catalysts.

[27–29]. However, nitrogen oxides contribute little to the catalytic activity of carbon catalysts [15]. SA Fe-g-C₃N₄ (600) exhibited the highest relative content of pyridinic N in N species among all SA Fe-g-C₃N₄ samples, further confirming that the pyridinic N believed to serve as the coordinating sites for Fe species (Table S2). Furthermore, the pyridinic N's peak showed an obvious deviation, which can be closely related to the bonding of pyridinic N atoms with Fe atoms to form Fe-N_x sites [30]. The relative content of pyridinic N increased with the increase in pyrolysis temperature and promoted the forming of Fe-N_x sites. Therefore, the increase in pyrolysis temperature can increase the relative content of pyridinic N, and more importantly, more Fe-N_x species were formed as active centers.

To investigate the atomic local structure and coordination information of SA Fe-g-C₃N₄ (600), X-ray absorption fine structure (XAFS) and extended XAFS (EXAFS) measurements were performed. As shown in the X-ray absorption near-edge structure spectra of the Fe K edge for SA Fe-g-C₃N₄ (600) (Fig. 3a), the rising edge gradually moved to the higher energy region with the increase in element valence of Fe. The rising edge position of SA Fe-g-C₃N₄ (600) was between those of FeO and Fe₂O₃, indicating the existence of the unique electronic structure of Fe^{δ+} ($2 < \delta < 3$) in the samples. In Fig. S4, two obvious peaks were observed according to the Fourier transform (FT) EXAFS spectra, the dominant peak at ~ 2.1 Å could be assigned to the Fe-N_x scattering paths.

Another peak at ~ 4.2 Å was observed, but it was different from those of Fe foil, FeO, and Fe₂O₃. From the quantitative EXAFS fittings (Table S3), the coordination features of SA Fe-g-C₃N₄ (600) were determined to be 4.2 for Fe-N, indicating the existence of dominated single Fe atoms with the incorporation of SA Fe-g-C₃N₄ (600). The FT k³-weighted $\chi(k)$ function EXAFS spectra (Fig. 3b) of SA Fe-g-C₃N₄ (600) showed an Fe ligand peak at 2.52 Å, distinguishing the Fe-Fe coordination peak at 2.21 Å in the Fe foil and the Fe-O coordination peak at 2.70 Å in the FeO sample and 2.61 Å in the Fe₂O₃ sample,

indicating no obvious Fe-Fe bonding. This further evidenced the atomic dispersion of Fe in the SA Fe-g-C₃N₄ (600) catalyst. The dominant peak at ~ 1.47 Å could be attributed to the Fe-N coordination. Moreover, the WT contour plots of SA Fe-g-C₃N₄ (600) displayed one intensity maximum at ~ 4.1 Å⁻¹ (Fig. 3c). An intensity maximum at ~ 7.1 Å⁻¹ associated with the Fe-Fe connection was not observed in SA Fe-g-C₃N₄ (600), indicating that isolated Fe atoms were the dominated Fe species in the as-prepared catalyst [31,32]. To obtain the quantitative structural parameters of Fe in the SA Fe-g-C₃N₄(600) catalyst, the least square EXAFS curve fitting was performed (Fig. S5). In addition, the coordination features indicated that all Fe sites in SA Fe-g-C₃N₄ (600) can be designated to be both single Fe atoms and Fe atomic clusters. The results demonstrate that the Fe-N_x coordination number of the SA Fe-g-C₃N₄ (600) catalyst was ~ 4 , indicating that the isolated Fe atoms were four-fold coordinated with N atoms.

To confirm the coordination structure of Fe-N₄ in SA Fe-g-C₃N₄ (600), DFT calculations were performed to acquire the configurations of defective Fe-N_x-C. Because there is usually an incomplete structure in these cases and the actual incomplete condensation of g-C₃N₄, the actual component was modified to C₃N_{4.5}H_{1.5} using the results of element analysis, which is used to represent the g-C₃N₄ sheet [33]. Fig. 4 shows the DFT calculation details. Several possible doping sites of Fe atoms on a monolayer g-C₃N₄ sheet were investigated. As shown in Fig. 4b–f, as one H atom was removed, the Fe-N₄(1) configuration showed much lower formation energy (2.794 eV) than those of Fe-N₃ (4.746 eV) and Fe-N₄ (2) (4.113 eV). This result indicates that the isolated Fe atoms tend to occupy the four-fold coordinated doping configuration (Fig. 4d), which is consistent with EXAFS fitting results.

3.2. Catalytic degradation activity of the synthesized catalysts

The adsorption and catalytic performances of the catalysts prepared at different pyrolysis temperatures (400 °C, 500 °C, and 600 °C) were

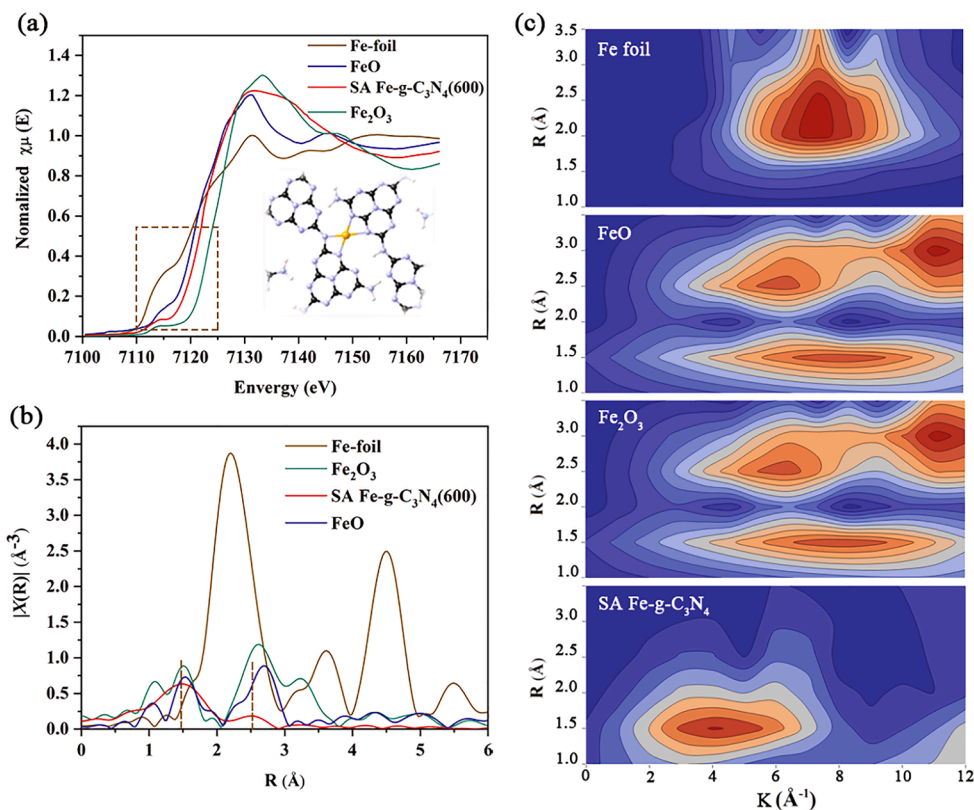


Fig. 3. (a) XANES spectra at the Fe K-edge of SA Fe-g-C₃N₄(600) and reference samples; (b) FT k³-weighted EXAFS spectra of SA Fe-g-C₃N₄ (600) and reference samples; (c) WT plot of SA Fe-g-C₃N₄ (600).

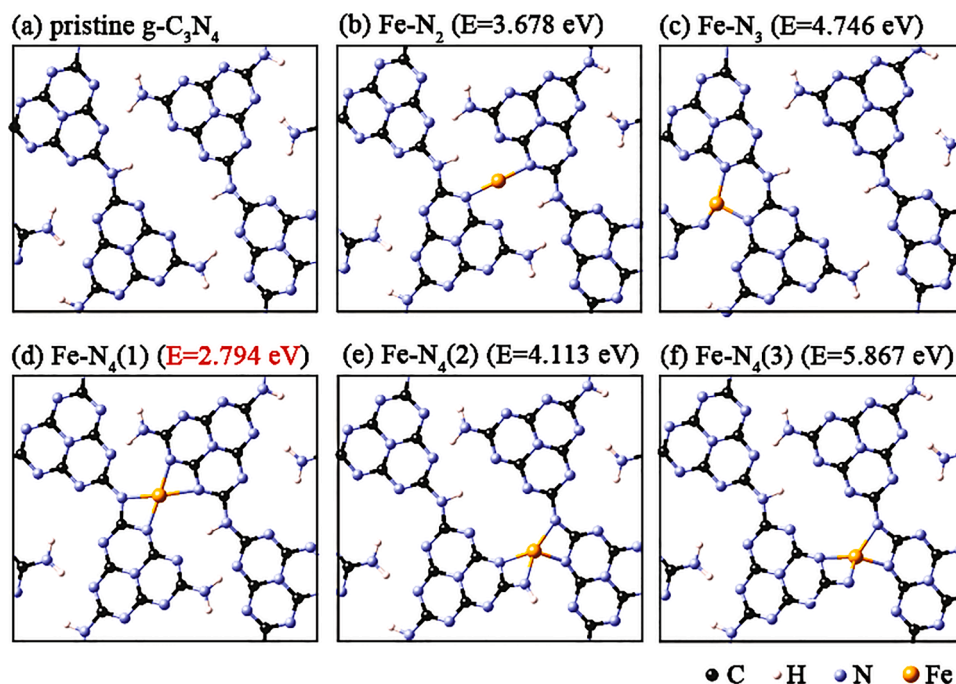


Fig. 4. Theoretical analysis of the $\text{Fe-N}_x\text{-C}$ structure. The schematic structure of (a) pristine $\text{g-C}_3\text{N}_4$ and (b-f) Fe-doped $\text{g-C}_3\text{N}_4$ with different doping configurations. The calculated formation energies (E) of Fe-doped $\text{g-C}_3\text{N}_4$ with different doping configurations are also given.

evaluated in Fenton-like reactions (Fig. 5a–b). Removing TC by adsorption increases from 1.68% to 29.53% with an increase in temperature, and the corresponding catalytic capacity increased from 77.52% to 93.29%. As evidenced by the Brunauer–Emmett–Teller specific surface area and pore size distribution of the synthesized catalysts (Table S4), SA Fe-g- C_3N_4 (600) exhibited a larger specific surface area, which was 5.47 and 15.82 times higher than those of SA Fe-g- C_3N_4 (500) and (400), thus providing more reactive sites for the degradation process. Furthermore, g- C_3N_4 prepared at 600 °C and Fe_3O_4 was also applied to TC degradation, respectively. Under the same reaction conditions, the TC degradation rates using g- C_3N_4 and Fe_3O_4 were only 30.6% and 29.4%, respectively. The inset in Fig. 5b shows a high linear correlation ($R^2 = 0.99$) of the corresponding first-order reaction rate constants (k_{obs}) and Fe content on the samples, indicating that more single Fe atoms could be coordinated with the 2-MI-derived N, forming more Fe-N_4 sites for PMS activation with increase in pyrolysis temperature. Therefore, the catalyst obtained at 600 °C exhibited a faster TC

removal than the catalyst pyrolyzed at lower pyrolysis temperatures. To ensure high TC removal efficiency, all SA Fe-g- C_3N_4 catalysts in the following sections were calcined at 600 °C.

When the PMS concentration increased from 0 mM to 0.25 mM and 0.50 mM, the TC removal efficiency enhanced from 43% to 93.29%, and the corresponding reaction rate constant increased 1.51 times (Fig. S6a). However, when the PMS concentration further increased to 0.75 mM and 1.00 mM, no considerable influence occurred on the TC degradation rate. This phenomenon can be explained as the concentration of the catalyst (0.1 g L^{-1}) was insufficient to provide enough active sites for activating a high PMS concentration. Therefore, the effect of catalyst dosage was further studied (Fig. S6b). The results show that the TC degradation followed first-order kinetics. TC removal can be significantly enhanced as the dosage of the catalyst increases. As shown in the inset of Fig. S6b, the TC degradation rate increased linearly as the dosage of SA Fe-g- C_3N_4 (600) increased from 0 to 0.20 g L^{-1} . The results showed that SA Fe-g- C_3N_4 (600) could provide more active sites

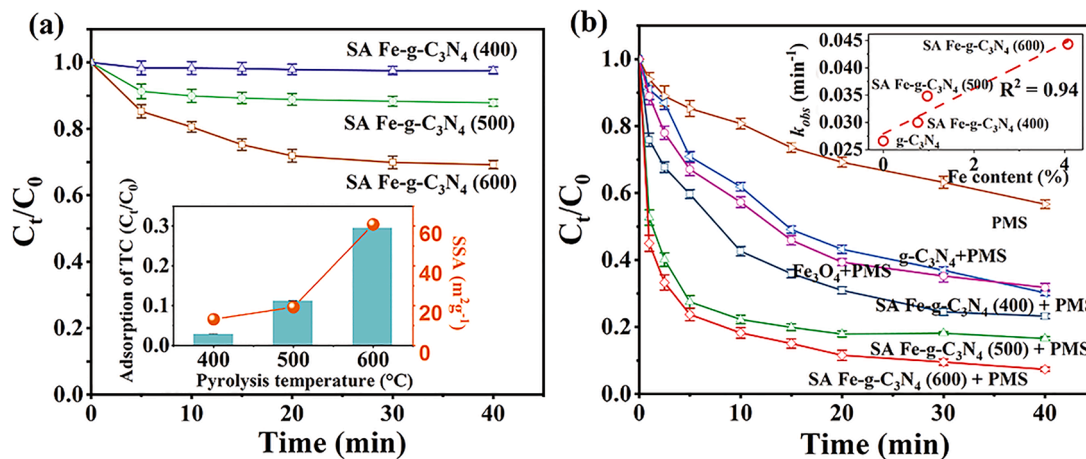


Fig. 5. (a) Adsorption and (b) catalytic properties of the catalysts prepared at different pyrolysis temperatures. Reaction conditions: TC = 10 g L^{-1} , catalyst dosage = 0.1 g L^{-1} , PMS concentration = 0.5 mM, initial pH = 6.5 and temperature = 25 °C.

for PMS activation and generate more active oxidation species for TC degradation.

Because TC has three pK_a values ($pK_{a1} = 3.3$, $pK_{a2} = 7.7$, and $pK_{a3} = 9.7$), the main distribution species of TC and the electrostatic interaction between TC and the catalyst will be affected by the solution's pH, affecting the catalytic degradation process [34]. Therefore, the influence of different initial pH on TC degradation using the SA Fe-g-C₃N₄ (600)/PMS system was explored. From Fig. S6c, the result indicates that both peracidity and alkalinity had adverse effects on TC degradation, and the optimum initial pH was weak acid. The TC degradation rate using the SA Fe-g-C₃N₄ (600)/PMS system differed under different initial pH and presented the following rank order: pH 6.5 > pH 6.0 > pH 7.0 > pH 9.0 > pH 5.0 > pH 3.0. The influence of pH on the catalytic oxidation process is mainly because of (i) the charge of the pollutant, catalyst, and oxidant, (ii) the existing form of the oxidant, and (iii) the mutual and type of transformation of reaction groups at different pH values [35]. The initial pH value of the solution would affect the transformation of ROS [36]. When the system is in an alkaline condition, the high-valence iron species ($Fe^{IV} = O$) can also be produced and reacted with organic anions, and PMS would undergo self-decomposition to HSO_5^{2-} , which is unable to generate $SO_4^{\bullet-}$. Thus, TC degradation in the SA Fe-g-C₃N₄ (600)/PMS system can proceed more favorably at near-neutral pH [37].

Moreover, the effect of temperature on the SA Fe-g-C₃N₄ (600)/PMS process was also evaluated (Fig. S6d). With an increase in temperature from 25 °C to 55 °C, the degradation efficiency increased because of the promotion of the transfer of active substances to TC at high temperatures. When the temperature continued to increase to 65 °C, the degradation rate accelerated in the first 5 min, but then decreased. This was because the PMS activation mechanism involved the fission of the O-O bond in the PMS structure and high temperature (greater than 50 °C), thus breaking the O-O bond in the PMS structure and accelerating the reaction [38]. However, an increase in temperature could also accelerate the side reaction, resulting in the degradation rate decrease after 5 min [39]. To obtain the highest TC removal efficiency and avoid wasting energy, 25 °C was selected as the best temperature.

3.3. Reactive species involved in TC degradation

To verify the active species in the reaction system, quenching experiments and 5,5-Dimethyl-1-pyrroline N-Oxide (DMPO)-trapped EPR experiments were conducted (Fig. 6). DMPO-trapped EPR experiments showed that DMPO- $\bullet OH$ and DMPO- $SO_4^{\bullet-}$ signals can be easily detected in SA Fe-g-C₃N₄ (600) suspension in the presence of PMS, indicating

$\bullet OH$ and $SO_4^{\bullet-}$ radicals in the reaction process. MeOH was used as a probe to detect the effect of $SO_4^{\bullet-}$ and $\bullet OH$ [14]. Furthermore, isopropyl alcohol (IPA) and *tert*-butanol (TBA) were used to prove the $\bullet OH$ contribution to the reaction. As shown in Fig. 6a and Fig. S7, adding 1 M MeOH, 0.5 M IPA, and 0.5 M TBA reduced the TC degradation rate by 20%, 12%, and 10%, respectively, verifying that $SO_4^{\bullet-}$ and $\bullet OH$ were not the main active substances in TC degradation using the SA Fe-g-C₃N₄ (600)/PMS system. Besides, DMPO- $O_2^{\bullet-}$ signals were also observed in EPR spectra (Fig. 6c). Adding p-BQ and hexachloroethane (scavengers for $O_2^{\bullet-}$ radicals) reduced the TC degradation rate by ~ 25% and 13%, respectively, indicating the indispensable roles of $O_2^{\bullet-}$ radicals in the catalytic reaction.

To verify the non-radical process, FFA was selected as a scavenger to quench 1O_2 . The TC degradation efficiency was significantly inhibited to 42% after adding FFA. Similar to N-doped graphene, the generation of 1O_2 could be because of the C atoms activating PMS in the g-C₃N₄ framework. Furthermore, the dissolved oxygen in water can also promote the formation of 1O_2 [35]. Therefore, the experiment of trapping 1O_2 was conducted in an N₂ atmosphere. The results show that the inhibition effect of FFA on TC degradation was reduced to 15% under anaerobic conditions (Fig. 6a). To further explore the formation of 1O_2 in the SA Fe-g-C₃N₄ (600)/PMS system, the EPR experiment was conducted using 2,2,6,6-tetramethylpiperidine (TEMP) as a spin-trapping agent. As shown in Fig. 6d, the strong 1:1:1 triplet signal was quickly observed when TEMP was added to the SA Fe-g-C₃N₄ (600)/PMS system, which further proved the presence of 1O_2 , but reactive species other than 1O_2 existed.

The charge transfer among catalysts, PMS, and TC molecules were determined by electrochemical evaluation. In Fig. S8a, injecting PMS and TC significantly increases the current, indicating the generation of current flow from organics to the surface of SA Fe-g-C₃N₄ (600) [40]. The current densities of SA Fe-g-C₃N₄ (600)/PMS/TC and SA Fe-g-C₃N₄ (600)/PMS systems were higher than that of the bare SA Fe-g-C₃N₄ (600) system (Fig. S8b) because of the formation of meta-stable reaction complexes between the PMS molecules and active center [41]. The electrochemical impedance spectroscopy Nyquist plots of SA Fe-g-C₃N₄ (600) were smaller than those of other catalysts and reveal its significantly lower charge transfer resistance and better electron transfer process than other catalysts (Fig. S8c).

Recent studies have revealed that high-valent iron-oxo species, which are dominant reactive species, can be formed by the oxidation of the coordination of Fe and N-based ligands using PMS [42]. Thus, TC degradation in the SA Fe-g-C₃N₄ (600)/PMS system can involve high-

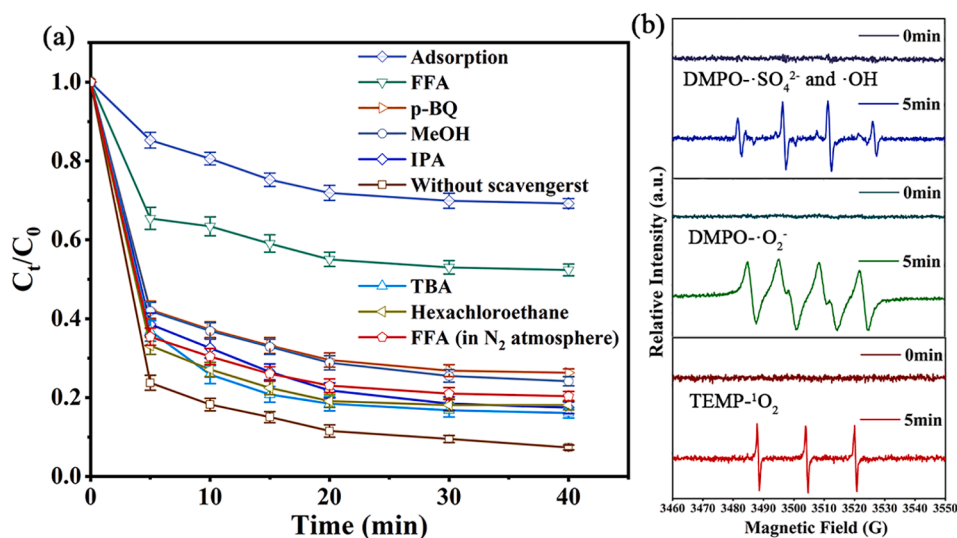
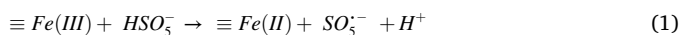


Fig. 6. (a) The effects of different scavengers on TC degradation by SA Fe-g-C₃N₄ (600)/PMS system and the corresponding rate constants; (b-c) DMPO-trapped EPR spectra of $SO_4^{\bullet-}$, $\bullet OH$ and $\bullet O_2^{\bullet-}$ and (d) TEMP-trapped EPR spectra of 1O_2 .

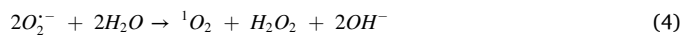
valent iron oxide. To verify this assumption, DMSO, a scavenger of high-valent iron-oxo species that can be oxidized by high-valent iron to form dimethyl sulfone (DMSO₂), was added to the reaction system. Subsequently, TC degradation significantly decreased from 93.3% to 31.0% in 40 min. The strong inhibition could be because of the competition between DMSO and TC for high-valent iron-oxo species. The high-valent iron-oxo species were considered the main reactive species for SA Fe-g-C₃N₄ (600) to activate PMS. Therefore, the degradation intermediates after adding DMSO were analyzed using GC-MS. As shown in Fig. S9, DMSO₂ was detected in the solution after the reaction, further proving the existence of high-valent iron-oxo species [43]. Furthermore, citrate was added to the system, which combined with Fe sites to inhibit the formation of high-valent iron-oxo species, verifying the contribution of high-valent iron-oxo species to TC degradation [23,42]. As shown in Fig. S10, TC degradation was significantly inhibited by adding citrate, indicating that high-valent iron-oxo species play a leading role in TC degradation in this system. This result confirms that the high-valent iron-oxo species functioned as the dominant active species in the SA Fe-g-C₃N₄ (600)/PMS system for TC degradation.

As shown in Fig. 7, a catalytic pathway of TC degradation by SA Fe-g-C₃N₄ (600)/PMS system was proposed. In the process of catalytic oxidation, because of the peroxy bond in the PMS molecule, the PMS bond combines with Fe-N₄ site and functions as an electron donor, which was decomposed to generate SO₅²⁻ (Eq. (1)). The produced Fe(II) could activate PMS to form the Fe(III) and SO₄^{•-} (Eq. (2)) [12,44,45]. Simultaneously, •OH was produced by the reaction between SO₄^{•-} and H₂O (Eq. (3)).

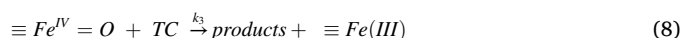
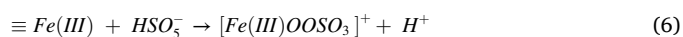


Studies on degrading pollutants using activated PMS indicated that there were multiple ways to produce ¹O₂. Both the self-oxidation of PMS, the combination of PMS with carbonyl moiety on carbonaceous materials or the carbonyl groups and the evolution of O₂^{•-} can lead to the production of (see Eq. (5)) [46]. In the SA Fe-g-C₃N₄ (600)/PMS system, the formation of ¹O₂ could correlate with the direct self-oxidation of PMS Eq. (5), the interaction between g-C₃N₄ and PMS, or the recombination of O₂^{•-}. To verify this assumption, the degradation and acquisition experiments of g-C₃N₄/PMS and Fe₃O₄/PMS systems were also conducted. As shown in Fig. S11, the active species in the g-C₃N₄/

PMS system was dominated by ¹O₂, whereas the Fe₃O₄/PMS and only PMS systems were dominated by •OH and SO₄^{•-}. It was reasonable to speculate that the main species ¹O₂ was produced in the SA Fe-g-C₃N₄ (600)/PMS system because of the combination of PMS and g-C₃N₄. Furthermore, the degradation rate decreased by ~25% when p-BQ was added, indicating that ¹O₂ production was related to O₂^{•-}.



Moreover, the Fe(III) in the Fe-N₄ center can also be coordinated with the O atom in PMS to form Fe(III)OOSO₃ complexes (Eq. (6)), whereas the heterolysis of O-O in Fe(III)OOSO₃ complexes results in high-valent iron-oxo species, Fe^{IV}=O (Eq. (7)) [12]. TC and intermediate products were attacked by the produced Fe^{IV}=O to form small mineralized molecules and Fe^{IV}=O was decomposed into Fe(III) (Eq. (8)) [47]. Therefore, the nonradical process controlled by Fe-N₄ sites can be critical in the degradation process, which mainly decomposes organic pollutants into small molecules through high-valent iron-oxo species. To further verify the main role of Fe-N₄ sites, oxalate and SCN⁻ were added to the reaction system, which can form a metal carboxyl complex to block the Fe sites, and quench the reaction strongly [48]. Fig. 8a-b show that TC degradation was significantly inhibited with the increasing concentration of oxalate and NaSCN, suggesting the contribution of Fe sites to the activation. The Fe(III) species in the atomically dispersed Fe-N₄ sites could be coordinated with the O atom in the peroxide bond of PMS activation to form high-valent iron-oxo species. Thus, the adsorbed TC was oxidized by the adjacent Fe^{IV}=O, which can be reconverted to Fe(III) species. Therefore, TC degradation using SA Fe-g-C₃N₄ (600)-activated PMS was through the nonradical pathway, and the high-valent iron-oxo species were the major ROS.



3.4. Mechanism for PMS activation by SA Fe-g-C₃N₄ (600)

Selected pollutant degradation using g-C₃N₄/PMS, Fe₃O₄/PMS, SA Fe-g-C₃N₄ (400)/PMS, SA Fe-g-C₃N₄ (500)/PMS, and SA Fe-g-C₃N₄

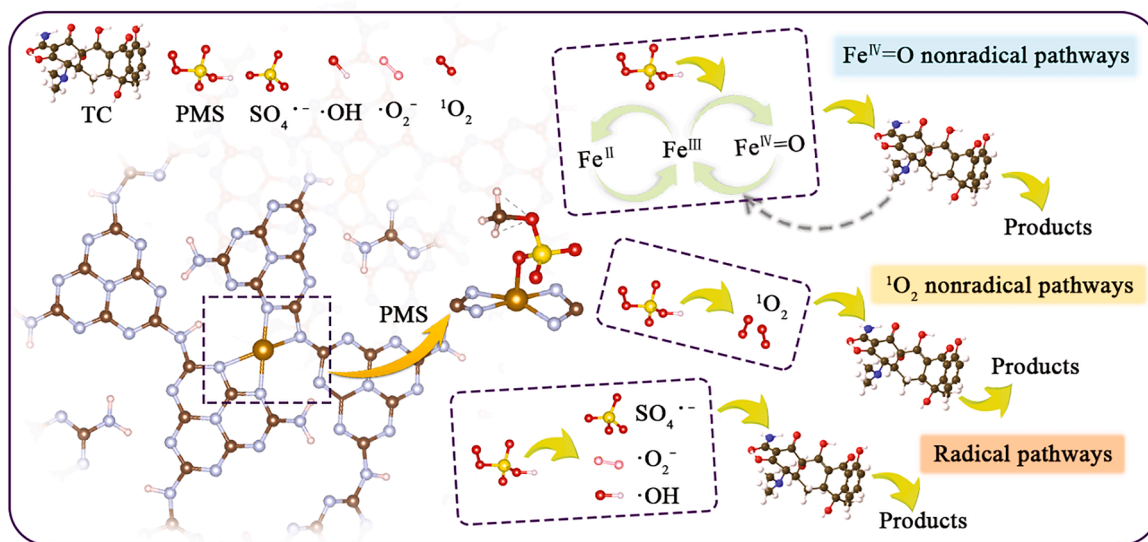


Fig. 7. The proposed mechanism for the Fenton-like reactions in the SA Fe-g-C₃N₄ (600)/PMS system.

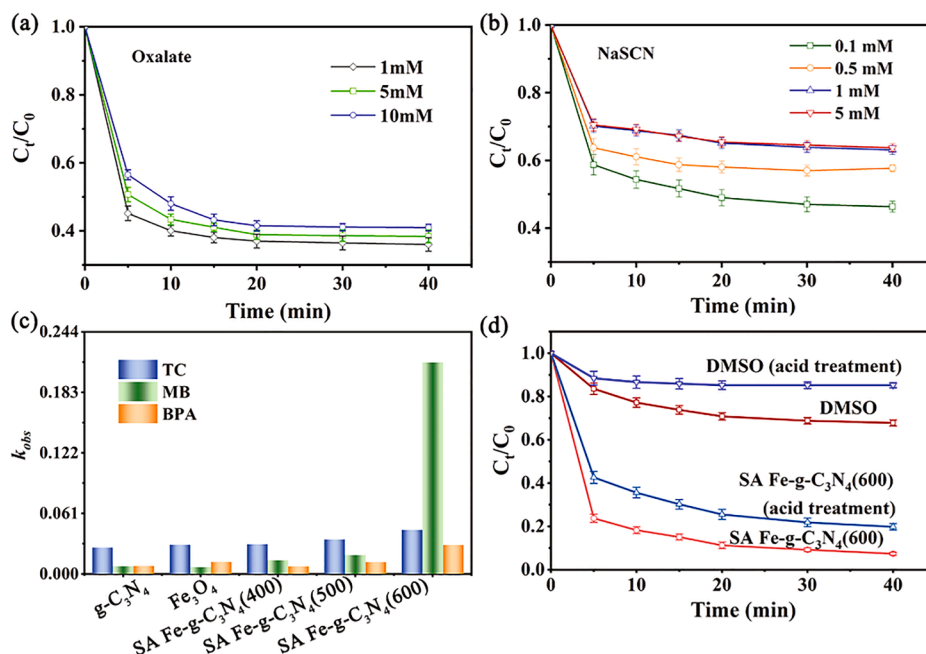


Fig. 8. (a) The effects of oxalate and citrate on TC degradation by SA Fe-g-C₃N₄ (600)/PMS system; (b) k_{obs} of different pollutants by different catalysts; (c) the catalytic performance of SA Fe-g-C₃N₄ (600) after hot acid-treatment and the quenching effect by DMSO. (PMS: 0.5 mM; catalyst: 0.1 g/L; pH: 6.5; TC concentration: 10 mg L⁻¹).

(600)/PMS systems was performed (Fig. 8c and Fig. S12). g-C₃N₄ exhibited low catalytic activity for selected pollutant degradation, indicating that both C and N atoms of g-C₃N₄ were not the main active sites for TC degradation [42]. Note that the removal effect of Fe₃O₄/PMS systems on selected pollutants was also poor. However, the SA Fe-g-C₃N₄ (600)/PMS system exhibited remarkably high degradation rates for several pollutants through PMS activation compared with other catalytic systems, indicating that the single Fe atoms in the structure of SA Fe-g-C₃N₄ (600) catalyst served as active sites for pollutant

degradation. To explore the role of different Fe species of SA Fe-g-C₃N₄ (600) in catalytic TC degradation, acid etching treatment was performed to remove most Fe clusters from the catalyst surface and only single Fe atoms remained evenly distributed on the g-C₃N₄ framework [49,50]. Therefore, SA Fe-g-C₃N₄ (600) was acid-treated to remove Fe metal clusters with 5 M H₂SO₄. As shown in Fig. S13a and S13b, the AC-HAADF-STEM images of SA Fe-g-C₃N₄ (600) obtained before and after heating acid treatment showed that most Fe clusters were removed, and only single Fe atoms remained on the g-C₃N₄ framework. The TC

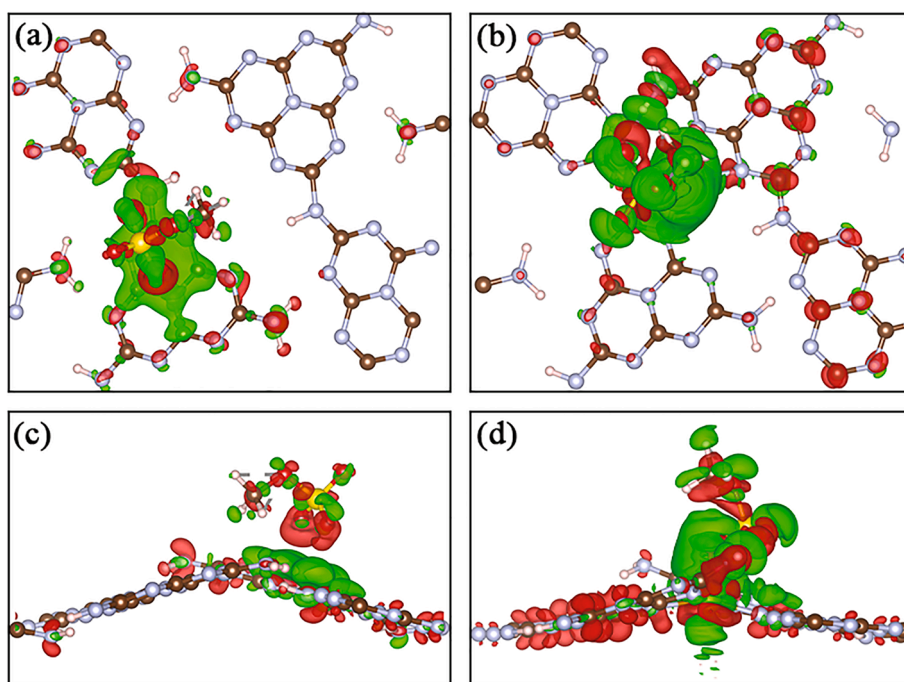


Fig. 9. Top view and side view of the charge density difference for optimized structure model of PMS adsorbed on (a, c) g-C₃N₄ and (b, d) SA Fe-g-C₃N₄(600), respectively. The green regions illustrate charge depletion while the red regions indicate charge accumulation.

degradation experiment using SA Fe-g-C₃N₄ (600) was conducted after acid etching treatment (Fig. 8d).

The TC degradation performance using SA Fe-g-C₃N₄ (600) was only slightly reduced, although most Fe clusters were removed during acid etching treatment. It was inferred that the SA Fe-g-C₃N₄ (600) catalyst exhibited exceptionally high catalytic activity and rates for the TC via PMS activation compared with a comparatively large proportion of Fe clusters in catalytic systems. These results further indicate that the SA Fe-g-C₃N₄ (600) catalyst structure served as the main active sites for TC degradation. Furthermore, the DMSO quenching result indicates that acid-treated SA Fe-g-C₃N₄ (600) produced a large number of high-valence iron-oxo species in TC degradation, proving that the single Fe atoms were dominant in the catalytic oxidation reaction. For comparison, the atomic Fe clusters occupied most Fe species in SA Fe-g-C₃N₄ (600), but has a minor role in TC degradation.

DFT calculations are performed to distinguish the interaction among PMS, g-C₃N₄, and Fe-N₄ sites. Fig. 9 and Fig. S14 show the optimized configurations and top and side views of charge density differences of PMS on g-C₃N₄ and SA Fe-g-C₃N₄ (600). The g-C₃N₄ catalyst model observed a charge between N and C atoms. For the SA Fe-g-C₃N₄ (600) catalyst, most charge transfer between Fe and N was observed, whereas no obvious charge accumulation occurred between Fe and C, revealing the weak interaction between Fe and C. In addition, the observation of the charge accumulation between N and C was stronger than that of g-C₃N₄. This intensified charge redistribution was expected to produce more catalytic active sites around the single-atom Fe sites, thus achieving efficient TC degradation. Furthermore, the charge density difference indicates a significant electron transfer process between the PMS and Fe-N₄ sites, revealing that the intermediate interaction between the PMS and Fe-N₄ sites resulted in the efficient activation of the PMS system.

The relationship between the Fe content and degradation efficiency was further explored in the cyclic experiment using XPS analysis. The Fe content in the catalyst and k_{obs} decreased with the increase in using

times (Fig. S15a). A higher linear correlation ($R^2 = 0.9975$) occurred between Fe content and k_{obs} , proving that the high Fe content caused effective TC degradation. The role of N species and contents on the catalytic activity of PMS were further investigated using SA Fe-g-C₃N₄ (600). Fig. 10a shows the high-resolution XPS spectra of N1s of SA Fe-g-C₃N₄ (600) catalysts before and after catalytic reaction, reuse, and regeneration. Fig. 10b and Table S5 show the N and different N species content of fresh and used catalysts. The linear correlation ($R^2 = 0.97$) between pyridinic N content (wt.%) and Fe content (wt.%) confirmed the strong binding of pyridinic N to Fe atoms (Fig. S15b). Some pyridinic N atoms are bound to Fe atoms to form Fe-N_x species, which could be identified as the active centers for catalytic TC degradation. These Fe-N_x species existed in the form of Fe-N₄ with optimal binding energy according to the DFT results. In the second cycle experiment, the degradation efficiency was reduced by 3.3% compared with the fresh catalyst, whereas a decrease in the pyridinic N content from 33.39% to 29.65% occurred, which originated from the protonation of pyridinic N into N-H during the Fenton-like process. The decrease in pyridinic N could result in reduced adsorption sites of PMS and pollutants [51], which could decrease degradation efficiency. In the third and fourth cycle experiments, the content of pyridinic N had a similar trend. The catalytic activity of the catalyst could be well recovered via a simple thermal treatment to remove the surface adsorbates. Simultaneously, the content of pyridinic N was increased as expected. The results show that the higher the N content in the catalyst, the more pyridinic N transferred in the g-C₃N₄ layer, which was conducive to anchoring Fe atoms. Furthermore, the relationship between Fe and pyridinic N and the role of Fe-N species was further confirmed by the linear correlation ($R^2 = 0.9769$) of pyridinic N contents (wt.%) and k_{obs} in multiple reused samples (Fig. 10c). Therefore, it was attributed to the higher Fe-N species and anchored single Fe atoms content that facilitate TC degradation. Furthermore, the dried and reused catalysts at different temperatures were explored. The pyridinic N content of materials dried at 70 °C was higher than those dried at 60 °C (Fig. S16), which could be the

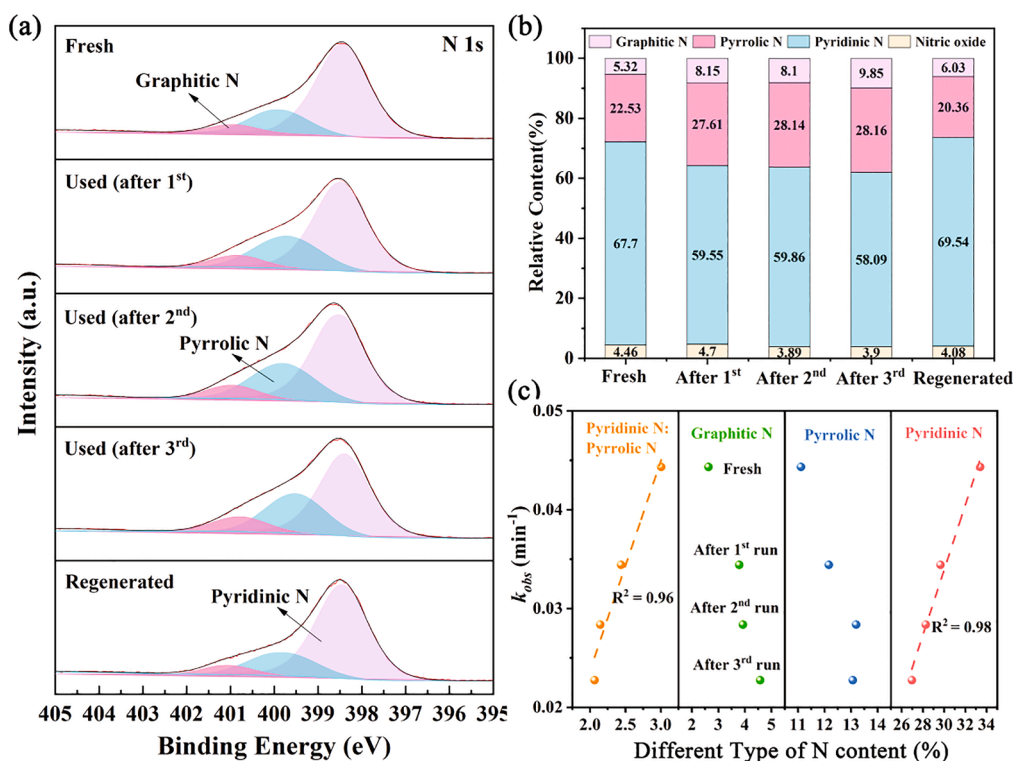


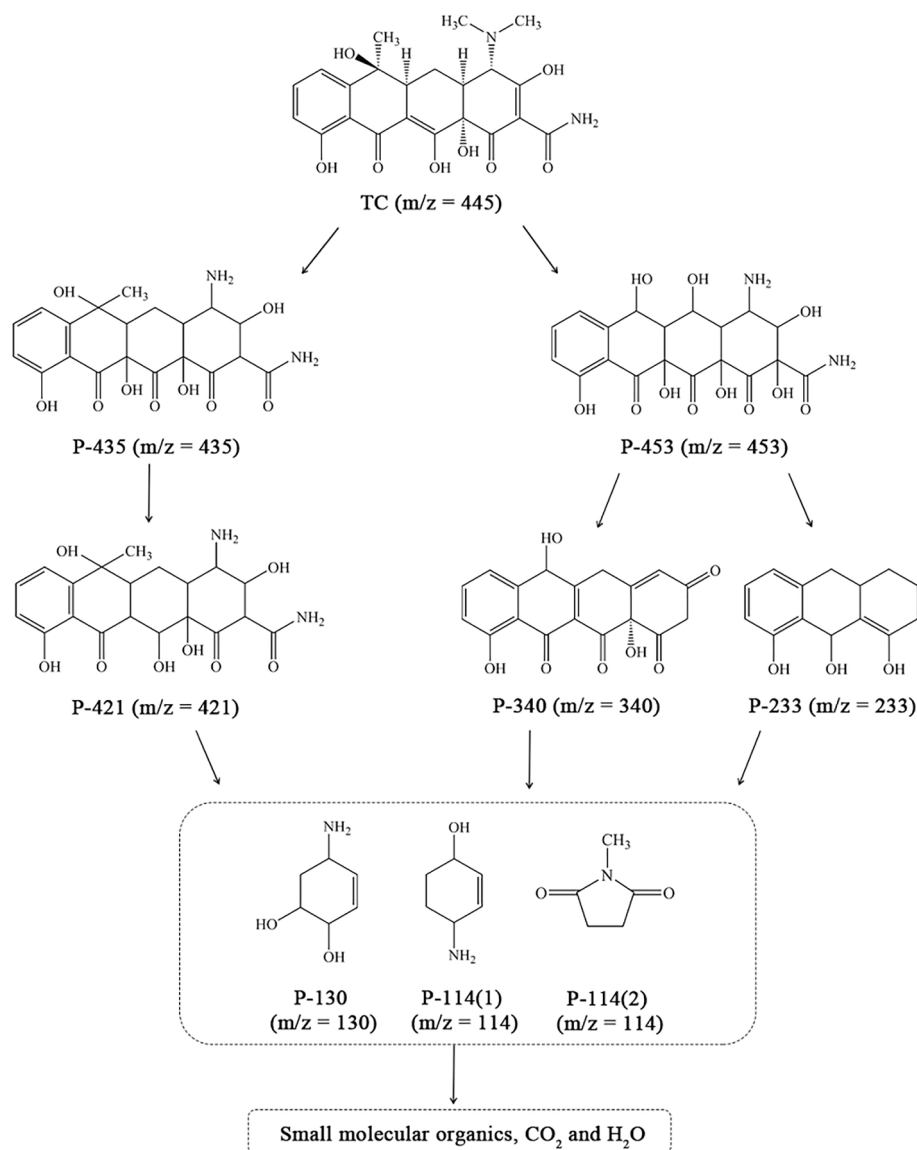
Fig. 10. (a) XPS spectra of N1s of fresh, reused and regenerated SA Fe-g-C₃N₄ (600) catalysts; (b) the related different N species content changes of SA Fe-g-C₃N₄(600) catalysts in batch reaction, and (c) the correlation of the pseudo-first-order kinetic constants of TC removal by SA Fe-g-C₃N₄ (600)/PMS system to different nitrogen functional group contents of SA Fe-g-C₃N₄ (600).

reason for the stronger catalytic performance of catalysts dried at 70 °C.

To reveal the proposed TC degradation pathway using the SA Fe-g-C₃N₄ (600)/PMS system, the intermediate products during the degradation process were determined using LC-MS at different reaction times. From the results, two degradation pathways of TC using an SA Fe-g-C₃N₄ (600)/PMS system were proposed (Scheme 1). The electron-rich groups in the TC molecular structure can react with active substances to form intermediates, especially the double bond in the TC structure. In pathway A, P-435 was generated by the loss of N-methyl, the addition of hydroxyl and hydrogen on the C-C double bond. This N-dealkylation occurred because of the low bond energy of N-C eroded by the SO₄^{•−} oxidation process [39]. P-421 was derived from the decomposition of P-435 through the dehydroxylation process, which is similar to 4-demethyltetracycline [52]. In pathway B, forming products P-453 was attributed to electron reduction by demethylation. Then, the intermediate P-453 was attacked by ROS generated in the SA Fe-g-C₃N₄ (600)/PMS system and transformed into P-340 (*m/z* = 340) and P-233 (*m/z* = 233) via ring-opening processes. Finally, some intermediates were further oxidized into small molecular organics (CO₂ and H₂O) because of the open-ring reactions and loss of some functional groups.

3.5. Stability of the SA Fe-g-C₃N₄ (600) catalyst

The stability and reusability of the heterogeneous SA Fe-g-C₃N₄(600) catalyst for long-term operations were crucial aspects in actual application, which were investigated via a continuous batch experiment under the same reaction conditions. As shown in Fig. 11a and Fig. S17, a high TC removal rate was observed during continuous batch reactions. TC was degraded by more than 90% in the first and second batch experiments. After four cycles, the TC degradation rate decreased by ~ 12%, and *k*_{obs} values decreased from 0.06753 to 0.06196 min^{−1}. The catalyst was washed three to four times with deionized water, separated, and dried (at 60 °C or 70 °C) after each use, and it was found that the performance of the catalyst dried at 70 °C was better than that at 60 °C. The negligible impact on the decrease in catalytic activity could be because of PMS adsorption or degradation products on the SA Fe-g-C₃N₄ (600) surface [40], or the Fe leached. A heating regeneration (350 °C for 3 h under N₂ atmosphere) was conducted after four cycles, and the catalytic activity of the used catalysts was mostly recovered. The recovery of the catalytic activity of SA Fe-g-C₃N₄ (600) could be because of removing intermediate products adsorbed on the active center and the edge of the catalyst after



Scheme 1. Proposed degradation pathway of TC in the SA Fe-g-C₃N₄(600)/PMS system.

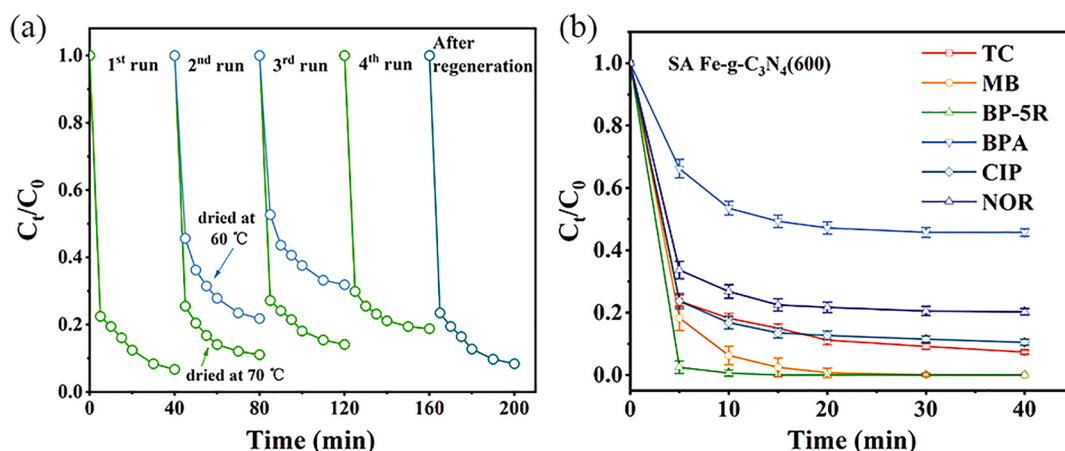


Fig. 11. (a) Cycling test for SA Fe-g-C₃N₄ (600). The catalyst was washed 4 times with deionized water after each cycle and then dried at 60 or 70 °C. If there is no remark, it should be dried at 70 °C; (b) removal efficiency of different organic pollutants using SA Fe-g-C₃N₄ (600) catalyst. Reaction conditions: initial concentration of pollutants = 10 mg L⁻¹, [PMS]₀ = 0.1 g L⁻¹, [catalyst]₀ = 0.1 g L⁻¹, initial pollution pH = 6.5 (without buffer solution), T = 298 K.

calcination [8,53]. The Fe leaching amount of SA Fe-g-C₃N₄ (600) in the TC degradation process was measured using ICP-MS. After the first catalytic oxidation reaction of SA Fe-g-C₃N₄ (600), the leaching iron in the reaction solution reached 0.02 mg L⁻¹, which was below the environmental standard (2 mg L⁻¹) [54]. The high activity and stability of the SA Fe-g-C₃N₄ (600) catalyst can be attributed to the high dispersion of Fe-N₄ sites in the carbon framework, which minimized the leached Fe atoms. Consequently, SA Fe-g-C₃N₄ (600) could maintain a high catalytic activity for TC degradation by activated PMS. Furthermore, Fig. 11b shows the degradation effect of different pollutants. SA Fe-g-C₃N₄ (600) catalysts exhibited good degradation efficiency on other antibiotics, such as CIP and NOR. Fast degradation of BP-5R and MB was also achieved by the SA Fe-g-C₃N₄ (600) catalyst via PMS activation, and the degradation rate of BP-5R reached 100% in 10 min, whereas only ~60% of BPA was degraded within 40 min.

To investigate realistic conditions, TC degradation using the SA Fe-g-C₃N₄(600)/PMS system in different water matrices (lake water, which was extracted from Kongmu lake, East China Jiaotong University, surface water and taping water) was studied. In Fig. S18a, the TC degradation rates in tap water, surface water, and lake water were 92%, 85%, and 81%, respectively. The reason for inhibiting TC degradation in lake water could be that the high concentration of organic matter or inorganic anions in water, which has covered the active sites of the catalyst or consumed the active substances, limits the catalytic performance of SA Fe-g-C₃N₄(600). To further investigate the effect of water characteristics on TC degradation using the SA Fe-g-C₃N₄(600)/PMS system, humic acid (HA, a natural organic matter) and some anions, including Cl⁻, HCO₃⁻, NO₃⁻, and H₂PO₄⁻ were introduced into the reaction system. In Fig. S18b–d, the HA and coexisting anions of NO₃⁻ and Cl⁻ had negligible adverse effects on TC degradation efficiency. The effect of Cl⁻ and NO₃⁻ anions on the degradation process is proportional to the reaction rate of the active radicals (•OH and SO₄⁻) [39], but the nonradical reaction is dominant in this experimental system. The nonradical-dominated oxidation systems cannot oxidize Cl⁻ [46]. Therefore, Cl⁻ hardly interferes with TC oxidation. When the concentration of HCO₃⁻ was lower, no significant influence on TC degradation occurred. However, the degradation efficiency was inhibited at a higher HCO₃⁻ concentration because when the concentration of HCO₃⁻ is high enough, it will have a negative effect on ¹O₂ [4]. Natural organic matter is ubiquitous in real water matrices. When the HA concentration was 5 mg L⁻¹, TC degradation was slightly affected, and the inhibiting effect did not further enhance with higher HA concentrations (Fig. S18e), which could be because of enough active centers on the surface of the catalyst, and weak reaction activity of HA with high-valent iron-oxo species and ¹O₂ [26]. Furthermore, with the increase in H₂PO₄⁻

background concentration, the TC degradation rate decreased from 93.29% to 75.58% (Fig. S18f), which could be because of the surface deactivation of the catalyst caused by H₂PO₄⁻ [55]. The above experiments further show that the degradation process of the system is dominated by nonradicals and the SA Fe-g-C₃N₄(600) could maintain high catalytic performance for TC degradation.

4. Conclusions

In summary, the SA Fe-g-C₃N₄ (600) catalyst exhibited exceptionally high catalytic activity (0.044 min⁻¹) for TC via PMS activation compared with other catalytic systems in this study. The highest catalytic oxidation performance of SA Fe-g-C₃N₄ (600) was associated with the different coordination structures of Fe-N_x and the interaction between different catalytic sites and PMS. The SA Fe-g-C₃N₄ (600) synthesized in this study provided abundant Fe-N₄ active sites with optimal binding energy for PMS activation to produce high-valent iron-oxo species for TC degradation. In the SA Fe-g-C₃N₄ (600)/PMS system, high-valent iron-oxo species and ¹O₂ played a dominant role in efficient TC catalytic oxidation degradation. The content of pyridinic N was positively correlated with TC degradation efficiency, guaranteeing the formation of more Fe-N₄ in SA Fe-g-C₃N₄ (600). The high stability of SA Fe-g-C₃N₄ (600) was verified in the cyclic reaction experiment. The catalytic activity of TC remained at a high level after four cycles and closed to the original level after regeneration, which could be attributed to the high dispersion of single Fe atoms served as Fe-N₄ active sites in the g-C₃N₄ matrices. This study makes a step forward for the formation of high-valent iron-oxo species in the process of PMS activation and will help develop and prepare other novel transition metal-based atom dispersion catalysts.

CRediT authorship contribution statement

Xiaoming Peng: Conceptualization, Methodology, Writing - review & editing. **Jianqun Wu:** Investigation, Methodology, Writing - original draft, Visualization. **Zilong Zhao:** Resources, Writing - review & editing. **Xing Wang:** Investigation, Methodology. **Hongling Dai:** Resources, Writing - review & editing, Supervision. **Li Xu:** Resources, Writing - review & editing. **Gaoping Xu:** Resources, Writing - review & editing. **Yan Jian:** Validation, Formal analysis, Visualization. **Fengping Hu:** Conceptualization, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.130803>.

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